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Making social bonds work at scale: what AI can offer public finance and investors

ESCP Impact Paper No.2025-16-EN

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Abstract

Social bonds aim to help governments raise capital for defined social priorities while offering investors transparent, outcome-linked instruments. Despite strong demand, their uptake remains limited due to administrative complexity, data fragmentation, and reporting burdens. This article examines whether artificial intelligence (AI) can support wider use of social bonds by easing these operational barriers.

Drawing on examples from public finance and current applications of AI, the paper explores how tools such as natural language processing and machine learning might assist with project selection, risk analysis, and reporting. It argues that AI can improve the usability and scalability of social bonds, provided its use remains transparent, accountable, and aligned with established standards. It also offers a practical implementation pathway for governments and investors, outlining how AI-supported processes can be introduced, disclosed, and scaled responsibly.

Keywords: Artificial Intelligence (AI), social bonds, public borrowing, fiscal transparency, public sector innovation

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Introduction

Governments today face growing pressure to finance social priorities while maintaining fiscal discipline and public trust. From housing and education to employment and health, public spending is expected to deliver measurable results under tighter resource constraints. At the same time, institutional investors are increasingly seeking opportunities to align their portfolios with social impact goals, while managing exposure to long-term policy and sustainability risks.

Social bonds were developed to match these two demands: offering a disciplined way for governments to raise capital for social objectives, and giving investors access to transparent, outcome-linked instruments. While the model is now well established, uptake has remained limited. The problem is not in the design, but in the operational burden of implementation, particularly for smaller or capacity-constrained public issuers, and for investors seeking consistent, comparable data.

This article explores how AI tools might support core processes in social bond issuance, including project screening, risk analysis, and reporting, and whether these tools can ease administrative demands without undermining the trust, judgement, and accountability that public finance requires. It outlines where AI tools can offer support, what conditions must be met to use them responsibly, and how implementation could be approached in practical terms by governments and investors.

What social bonds have offered and why they haven't scaled

Social bonds were introduced to help governments raise capital for defined social projects while providing investors with transparent, use-of-proceeds instruments linked to measurable impact. The structure is based on the Social Bond Principles developed by the International Capital Market Association (ICMA), which set expectations for project eligibility, use of funds, and reporting.

Since their introduction, social bonds have gained visibility and credibility through major public sector issuances, notably the European Union's SURE programme and France's Unédic. Development banks such as the Council of Europe Development Bank and the European Investment Bank have also played a role, alongside sovereign issuers in Spain, Italy, Germany, and the Netherlands, particularly during and after the COVID-19 crisis.

In our earlier ESCP Impact Paper (*Nawrot & Walkowicz, 2024*), we reviewed how social bonds offer distinct advantages for three key groups. For governments, they represent a more structured and credible borrowing instrument, reinforcing the link between financing and high-priority social needs. By requiring external reviews and regular updates, they can also support greater fiscal discipline. For investors, the clarity of the use-of-proceeds model and the transparency of reporting lower information asymmetry and support more reliable risk assessment. These features also help align investments with sustainability goals, making social bonds attractive to ESG-oriented portfolios. For the wider public, social bonds offer the possibility of more visible and accountable spending, where resources are tied to specific outcomes and populations.

Investor interest in social bonds has continued to grow, supported by steady market expansion. In 2024, global social bond issuance reached USD 251 billion, up from USD 159 billion in 2023. This made social bonds the fastest-growing segment within the labelled bond market that year (Battersby, 2025). According to a 2023 survey by Goldman Sachs Asset Management, cited by Reuters, 29% of the more than 700 investment professionals surveyed across 11 European markets already invested in the asset class, with a further 36% interested in allocating capital to it. Additionally, 45% of respondents planned to increase their investments in the next 12 months, while 42% said they would maintain them (Elisei, 2023).

As a most recent example showing the strength of the market, France's Unédic continued to attract significant investor interest through its ongoing social bond issuances, adhering to the ICMA Social Bond Principles. In March 2025, Unédic issued a €2 billion social bond with an 8-year maturity, backed by an explicit French state guarantee. The bond attracted strong investor interest, with orders exceeding €12 billion.¹ While the EU's SURE programme concluded its bond issuance phase in 2022, it remains a reference point for public sector social bond issuance due to its successful implementation and strong investor reception.

Despite this momentum, social bonds have not scaled beyond a relatively narrow group of issuers, largely due to operational demands such as eligibility assessments, third-party verification requirements, and limited data comparability. In 2024, green bonds accounted for USD 561 billion, or around 58% of total sustainable bond issuance. By contrast, social bonds made up USD 251 billion, representing just 26% of the market.² While both instruments follow the same use-of-proceeds structure set by ICMA, social bonds have proven more complex to apply in practice. Social spending tends to be more fragmented, often spread across different sectors such as housing, health, employment, or education, which makes it harder to bundle projects into large, clearly defined programmes. For public issuers, preparing eligible projects, fulfilling detailed reporting requirements, and coordinating third-party reviews place considerable demands on administrative capacity. For investors, limited standardisation across issuers makes it difficult to compare bonds or assess social impact at scale. As a result, social bonds, although widely recognised as credible, remain more complicated to issue for public authorities and more difficult to scale for investors seeking consistency and comparability across the market.

How AI could help scale social bonds

Although issuance has grown in recent years, social bonds are still used by a relatively small group of public issuers and have not yet become a routine instrument in either public borrowing or sustainable investment. The challenge lies not in the structure of the model, but in how it is applied, particularly given the operational constraints faced by public issuers and the information needs of investors. Many public issuers face practical constraints that limit their ability to use social bonds consistently. These include limited administrative capacity, fragmented data systems, and the burden of preparing, verifying, and communicating project-level information.

Artificial intelligence (AI) offers tools that could help address these operational bottlenecks by supporting key processes that currently limit their wider use.

¹ First Unédic successfully issued its first Social Bond of 2025, for an amount of €2bn, Investor News, Unédic, 2025, <https://www.unedic.org/en/investors/investors-news/first-Social-Bond-of-2025>

² Estimate based on: Battersby M. (2025), *Green, social, sustainability bond issuance reaches almost \$1trn*, Investment International, <https://investment-international.com/News/green-social-and-sustainability-bond-issuance-reach-almost-1trn/>

Governments and multilateral institutions are actively exploring the use of AI in public finance. In the United Kingdom, the Department for Work and Pensions has implemented AI tools to summarise large volumes of documents, improving efficiency in processing claims. The World Bank has developed the Procurement Anticorruption and Transparency (ProACT) platform, which uses AI to analyse procurement data in order to improve transparency and detect irregularities. In France, the government has launched initiatives such as the “Albert” AI system to modernise public services, including simplifying administrative procedures and improving accessibility for citizens. These examples illustrate the potential of AI to support complex administrative tasks in public finance across different institutional contexts.

While we have not found any publicly available evidence of AI being applied directly to social bond issuance, many of the processes involved, such as project screening, risk assessment, and reporting, appear well-suited to automation or data-supported tools. Social bonds, with their structured documentation and outcome-linked focus, are well-positioned to benefit from tools that increase consistency, reduce delays, and support comparability.

This section explores four areas where AI could help reduce the operational barriers to issuing and managing social bonds, along with the risks and trade-offs that must be considered from the outset.

1/ Identifying eligible projects

The process of selecting eligible projects is often a significant administrative task in social bond issuance, requiring time, judgement, and careful documentation. Governments must demonstrate that projects meet the criteria set out in their bond framework, typically aligned with ICMA’s Social Bond Principles, and show how each initiative addresses a defined social objective. This often involves reviewing a mix of programme documents, strategic plans, and local-level funding proposals, many of which are unstructured or inconsistently formatted, making it difficult to apply eligibility criteria systematically. AI tools, particularly those using natural language processing (NLP), a branch of machine learning focused on analysing text, could support this task by scanning documents and identifying projects that align with eligibility requirements. This would not replace public judgement, but could help governments screen options more efficiently and apply criteria more consistently.

For investors, a more structured approach to project selection improves transparency. When eligibility is applied in a systematic and evidence-based way, it becomes easier to understand what is being financed and to compare bonds across issuers, both of which support stronger confidence in the instrument.

However, there are limits to what AI can do in this area. Eligibility decisions often involve political priorities, local context, or timing considerations that may not be captured in the available data. If governments rely too heavily on AI tools, there is a risk that projects of social importance are overlooked simply because they are less well-documented or harder to classify. These tools should support, not define, the selection process.

2/ Assessing and communicating risk

Understanding and managing risk is essential for both issuing and investing in social bonds. For public issuers, this means anticipating potential delivery delays, cost overruns, or social outcomes that may fall short of expectations. For investors, it means having confidence that

material risks, whether related to policy changes, implementation failures, or weak project design, have been properly identified and addressed by the issuer.

AI tools, particularly those using machine learning, could support this work by analysing delivery records from previous programmes, surfacing patterns linked to delays, cost overruns, or underperformance. In this context, machine learning refers to algorithms that can detect patterns in structured data, such as delivery timelines or budget performance, to support project-level risk analysis. For example, delays in procurement, changes in policy, or weak partner capacity might be flagged as recurrent risk factors. Where such data exists, these tools could help issuers review proposed projects against known warning signs and strengthen early-stage scrutiny.

For investors, AI-generated insights may offer added value if they are transparent, comparable across issuers, and linked to established reporting frameworks. However, if models are based on incomplete or inconsistent data, they risk producing false confidence or obscuring judgement. Risk analysis supported by AI should inform, not replace, professional oversight and engagement between issuers and investors.

3/ Supporting monitoring and impact reporting

Once a social bond is issued, the issuer must report on how funds are allocated, which projects are supported, and what results are achieved. These reporting obligations are central to maintaining credibility with investors, but they are also one of the most resource-intensive parts of the process. Tracking outputs and outcomes often involves fragmented data, manual compilation, and coordination across different departments and reporting cycles.

AI tools could help ease this burden by automating elements of data extraction and consolidation. For example, structured financial data could be pulled from internal systems to produce draft reports on spending flows. Natural language processing could be used to extract outcome indicators from existing programme evaluations or qualitative field reports, reducing the need for manual review. If applied carefully, this could improve timeliness and internal consistency.

However, the usefulness of these tools depends on the structure and quality of the underlying data. Where inputs are inconsistent, AI is unlikely to produce reliable outputs. There is also a risk that automation leads to reporting that meets formal requirements but fails to convey whether funds were used effectively, or whether the reported outcomes reflect tangible social improvements. To support investor confidence and broader accountability, AI-generated reporting must be transparent, interpretable, and designed to inform real decision-making, not only satisfy formal compliance formats.

4/ Improving transparency for investors

Investor interest in social bonds depends not only on the quality of underlying projects but also on the clarity and regularity of information provided. Institutional investors often require comparable data across issuers, clear explanations of impact claims, and timely updates that fit within ESG and regulatory reporting cycles. Yet social bond communication practices remain uneven, with formats, metrics, and levels of detail varying widely.

AI could help improve consistency by generating standardised disclosures, visual summaries, and structured updates based on project and spending data. Dashboards could be designed to show progress against key indicators and highlight deviations from

expected timelines or outcomes. This could reduce the perceived complexity of social bonds and lower the entry threshold for investors less familiar with the asset class.

However, more communication is not always better communication. If AI-generated summaries prioritise visual appeal over substance, or if data is presented without clear context, these tools risk giving a false sense of precision. For investors to rely on AI-assisted communication, the outputs must be interpretable, traceable, and clearly aligned with the issuer's reporting framework. Tools should support understanding, not replace the need for scrutiny.

AI can help reduce several of the operational barriers that have limited the wider use of social bonds, from project selection and risk assessment to reporting and communication. For public issuers, this could mean greater efficiency, especially in contexts where capacity is limited. For investors, more consistent data could support comparability and better risk management. For the public, it may improve visibility and transparency. But these gains depend entirely on how the tools are used. Poor-quality inputs, models with unclear assumptions, or over-reliance on automation can undermine the credibility of the instrument. In the final section, we consider what responsible use of AI in social bonds might look like, and how tools can be designed to support, rather than distort, the core purpose of outcome-linked public borrowing.

Making AI work for social bonds

If AI is to contribute effectively to social bond markets, it must be applied in ways that solve real operational problems and support public objectives. It is about reducing the complexity that has made social bonds difficult to use, and doing so without weakening the standards that make them credible.

In practice, this means that any AI tool introduced into the social bond process should meet at least four basic conditions:

- **Problem-driven application:** The tool should respond to a clearly defined need, such as reducing duplication in reporting, identifying gaps in eligibility data, or aligning indicators across issuers. Tools without a targeted purpose are likely to distract rather than assist.
- **Transparency of logic and outputs:** AI-generated results must be open to scrutiny. Issuers and investors should be able to understand how a recommendation was reached, what data was used, and what assumptions were made.
- **Compatibility with public accountability:** Public borrowing decisions involve judgement, policy trade-offs, and social context. AI must support, not bypass, human responsibility in how funds are allocated and justified.
- **Alignment with existing standards:** Outputs should be consistent with recognised frameworks such as ICMA's Social Bond Principles. AI should improve adherence to these standards, not create parallel systems.

Turning AI's potential into operational practice will require clear, phased steps across both the public and investor-facing sides of social bond issuance. First, public issuers need to identify where existing administrative bottlenecks occur, such as in selecting eligible projects, tracking performance indicators across programmes, or consolidating data for reporting, and assess which parts of these workflows already rely on digitised records or follow repeatable structures. AI integration should begin in these zones of structured input, where tools like natural language processing can scan local project applications for alignment with eligibility criteria, or where machine learning can review financial and

outcome data for patterns linked to delay or underperformance. To make this practical, public finance ministries or subnational authorities could embed AI functions within the reporting templates or budget tracking systems they already use, without needing to build separate platforms. These tools could be developed internally by public digital agencies or procured from trusted public-sector vendors under clear contracts tied to recognised standards such as the ICMA Social Bond Principles or EU Taxonomy. Crucially, implementation must include protocols for human oversight and auditability, so that any automation supports accountability.

Second, the visibility of AI-supported processes must be built into the external communication strategy. Governments and public issuers would need to explain where AI is being used, such as in eligibility screening, data validation, or impact calculation, and how that use strengthens reliability. This could be disclosed in investor presentations, framework documents, reporting templates, or third-party assurance statements. Over time, such disclosure can help institutional investors understand not only the impact of a given bond, but the credibility of the processes behind it. Market-wide guidance bodies (such as ICMA, EU platform groups, or development finance institutions) could help by defining minimum transparency standards for AI use in labelled bonds.

Finally, adoption will depend on trust and interoperability. Governments must ensure that any AI tools introduced can work across ministries and datasets, and that local authorities can use them without excessive cost or technical knowledge. Pilot programmes, perhaps coordinated at the national or EU level, could be used to trial modular tools, e.g. an AI-powered eligibility checker or indicator dashboard, under controlled conditions. If designed to be modular, auditable, and standards-aligned, these tools could eventually support a more integrated infrastructure for outcome-based public borrowing. This would reduce the entry burden for smaller issuers while giving investors a more consistent view of risk and impact across the market. For investors, the value of AI lies in whether it improves the clarity, consistency, and comparability of information, not merely adds volume. Disclosures must remain explainable and aligned with recognised frameworks, so that automation strengthens confidence rather than undermines it.

Designed carefully and used well, AI tools can support both governments and investors in addressing the practical constraints that have limited the broader use of social bonds. They can help improve consistency, reduce duplication, and make key processes more manageable. But any improvement in efficiency must reinforce the reliability and purpose of the instrument. AI tools should enable wider adoption without lowering the quality of information or weakening the connection between public borrowing and the social outcomes it is meant to deliver.

This discussion also connects with wider debates on algorithmic governance and fiscal tech, which explore how AI is influencing not just operational processes but also the framing of accountability, transparency, and discretion in public finance. By focusing on a specific, underexamined use case, this paper adds a grounded contribution to those debates, showing how abstract principles of transparency and accountability play out in real public finance decisions.

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